

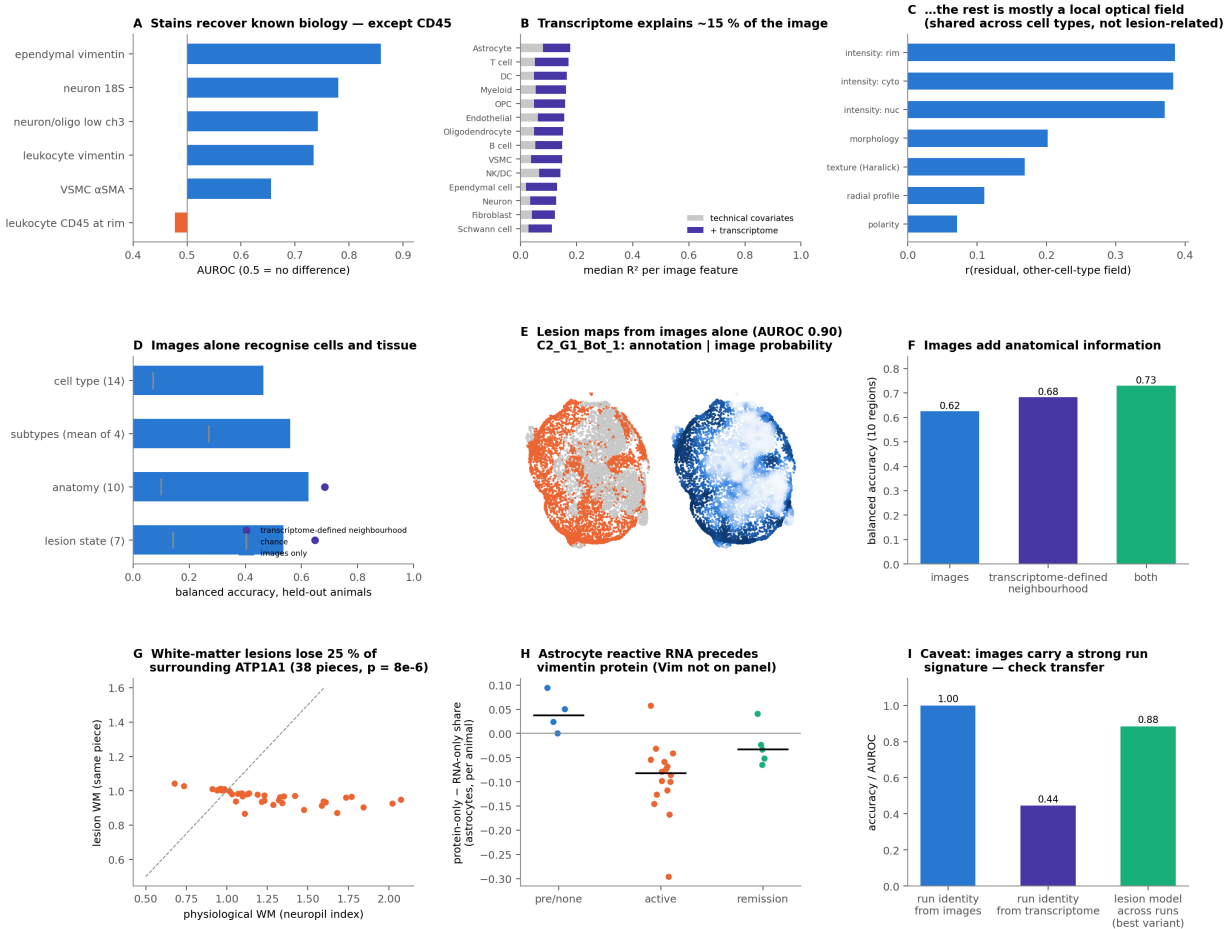
What does the Xenium *multimodal segmentation kit* tell us that the transcriptome doesn't?

Mouse EAE spinal cord, RRMAP2 runs 5 + 6 · 18 Xenium 5K images · 51 tissue pieces · 25 animals · 500,379 annotated cells · 4 stain channels · 201 image features per cell. Analyses in [notebooks/01–09](#) ; every number below comes from them.

The segmentation stains add *three kinds of information*: per-cell protein for genes the 5K panel doesn't contain, tissue context around each cell, and enough signal to recognise cells and map lesions from images alone, across imaging runs. They do not reveal hidden cell states.

- **Protein the panel can't measure.** *Vim*, *Acta2* and *Atp1a1* are not on the panel. The stains recover known biology in every animal and show that astrocyte reactive transcription precedes vimentin protein.
- **Tissue context.** White-matter lesions have 25 % less ATP1A1 neuropil around each cell ($p = 8 \times 10^{-6}$). Images add anatomical information to the transcriptome (0.68 → 0.73).
- **Images alone, held-out animals.** Cell type 46 % (14 types, chance 7 %). Lesion maps reach AUROC 0.90, and a model trained on one run maps lesions in the other at 0.87–0.90.
- **Not added.** CD45 is undetectable in mouse cord. For lesion identity and clinical score the images are informative but redundant with RNA. ~85 % of image variance isn't explained by the transcriptome, but that part is mostly a local optical/staining field, not biology.

Code: [src/beyondboundaries](#) · notebooks [notebooks/00–09](#) (paired percent scripts in [notebooks/py/](#)) · running log [RESULTS.md](#) · figures [results/<notebook>/](#) · full summary [report/BeyondBoundaries_summary.pdf](#) · lab-meeting deck [report/slides/](#).



source: notebook09_conclusions.ipynb

Figure 1 • The whole story in one figure. A, the stains recover known biology except CD45. B–C, the transcriptome explains ~15 % of each image feature, and the rest is mostly a local field shared across cell types. D–F, images alone recognise cells, map lesions and add anatomical information. G–H, biology only the images show: white-matter neuropil loss, and astrocyte RNA preceding vimentin protein. I, images carry a strong run signature, yet the lesion model transfers across runs (section 4).

1 Data and approach

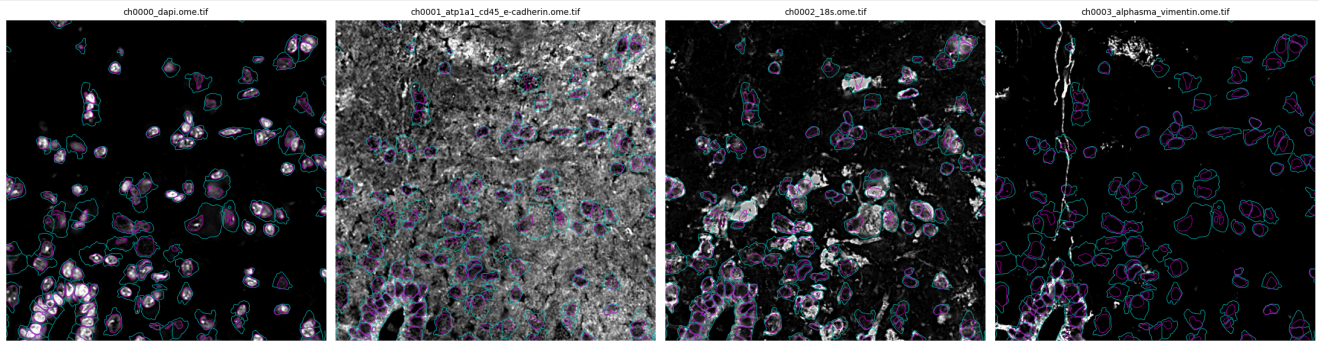


Figure 2 · The four segmentation-kit channels with Xenium cell (cyan) and nucleus (magenta) masks: DAPI · ATP1A1/CD45/E-cadherin (boundary) · 18S rRNA (interior) · αSMA/vimentin.

- **Data.** 25 Xenium bundles (XOA 4.0, fresh frozen, 0.2125 $\mu\text{m}/\text{px}$), 18 with annotation (500,379 cells; 19 L1 / 42 L2 cell types; curated lesion niches). Each image holds 2–3 tissue pieces from different animals. The statistical unit is the **animal**; cross-validation is grouped by animal throughout.
- **Features.** Per cell and channel: mean/percentile intensity in nucleus, cytoplasm, rim and an outer ring, territory intensity, Haralick texture, radial profile, polarity, plus shape (201 columns; mask areas match Xenium exactly; 0 truncated cells).
- **Normalisation.** Local background subtraction (σ 25 μm , cell-free tissue), then per image $(x - \text{bg}) / \text{p90}$ of physiological-niche cells. Per-piece scaling was rejected because piece intensity tracks disease.
- **Segmentation.** ~95 % of cells were segmented by 18S (interior), 0.5–4 % by the boundary stain, 2–3 % by nucleus expansion. Main analyses use 18S-segmented cells, with all cells as a sensitivity check.
- **Models.** Ridge (image $\leftrightarrow$ transcriptome, 50 PCs fitted in-fold), gradient-boosted trees (image-only prediction), leave-one-animal-out at animal level, and cross-run transfer (train run5 $\rightarrow$ test run6 and the reverse).

2 Do the stains measure what they claim?

Known biology is there in every animal; CD45 is not. VSMC αSMA (AUROC 0.66), leukocyte vimentin (0.73), low ch3 in neurons/oligodendrocytes (0.74), ependymal vimentin (0.86) and neuronal 18S (0.78) all hold in 25/25 animals and in boundary-segmented cells. Leukocytes are *not* brighter at the rim in the boundary channel (0.48), even where the neuropil is dimmest (0.41–0.49). In mouse spinal cord that channel behaves as ATP1A1 only.

Segmentation method shifts shape features strongly (boundary-segmented: outer radial bin +10 SD; nucleus expansion: solidity +6 SD). The between-type pattern is kept for 18S-segmented cells (r 0.998), hence the restriction.

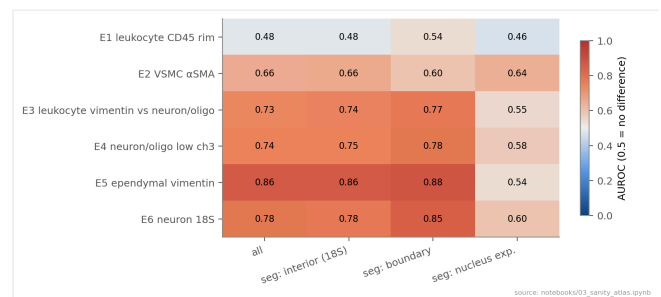


Figure 3 · Expectation tests per segmentation stratum (notebook 03).

3 How much of the image is new — and what is it?

The transcriptome explains little of the image. Median R^2 (covariates + transcriptome) per feature is 0.11–0.18 across 14 cell types (transcriptome-unique ΔR^2 0.06–0.10). Best explained are morphology (nucleus:cell ratio ΔR^2 0.40–0.47) and rim/cytoplasm intensity; least are polarity, radial profile and territory intensity.

The unexplained part is mostly optics. A cell's residual correlates with the mean residual of the 30 nearest *other-type* cells (rim/cyto/nuclear intensity r 0.37–0.39). This field follows local background (ATP1A1 glow into nuclei, $\rho = -0.55$), not lesion distance ($\rho \approx 0$). The cell-specific remainder shifts with lesion state in only 19/924 tests (~ 0.1 SD). Residual clustering finds no discrete hidden states.

In reverse, images predict transcriptional state. Neuron PC1 R^2 0.75 (neurofilament genes), astrocyte PC1 0.59 (*C3*, *Gfap* vs *Slc6a11*, *Aldoc*), oligodendrocyte PC1 0.43.

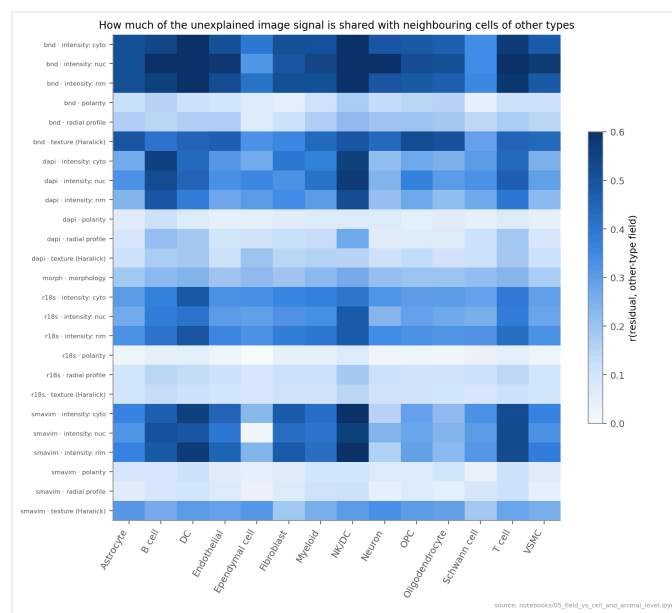


Figure 4 • Share of the unexplained image signal that is a cross-cell-type local field, per cell type × feature family (notebook 05).

What images alone can do

Cell type without a single transcript: 46 %

balanced accuracy over 14 types (chance 7 %; 44 % from the cell alone, 46 % with image neighbourhood). Ependymal 0.86, neuron 0.81 and Schwann 0.76; immune cells are poor (myeloid 0.16, T 0.26) because there's no CD45. Errors stay within lineage.

Subtypes: neurons 0.64 (chance 0.33; cholinergic 0.72), astrocytes 0.60 (reactive 0.68), myeloid 0.53 (microglia 0.72), oligodendrocyte lineage 0.47. Per stain, every channel carries identity (0.21–0.27 alone; all except 18S 0.40), so this isn't an artefact of the 18S masks.

Anatomy (10 regions): images 0.63 vs transcriptome cell-type composition 0.68; **both 0.73, so images add anatomical information.** Lesion state (7 classes): images 0.54, transcriptome 0.65, both 0.67.

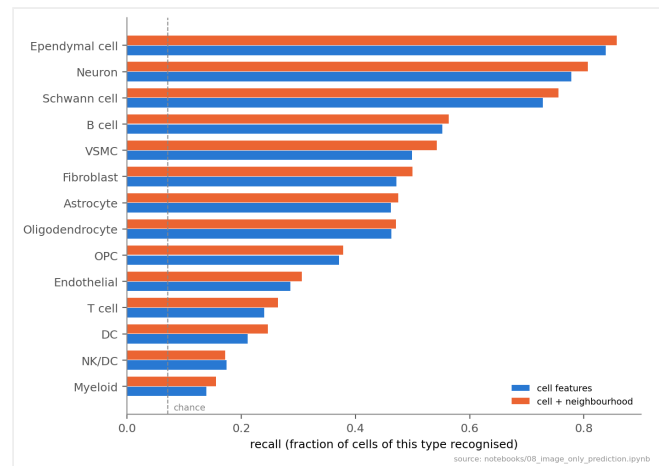


Figure 5 · Per-type recall from images only, held-out animals (notebook 08).

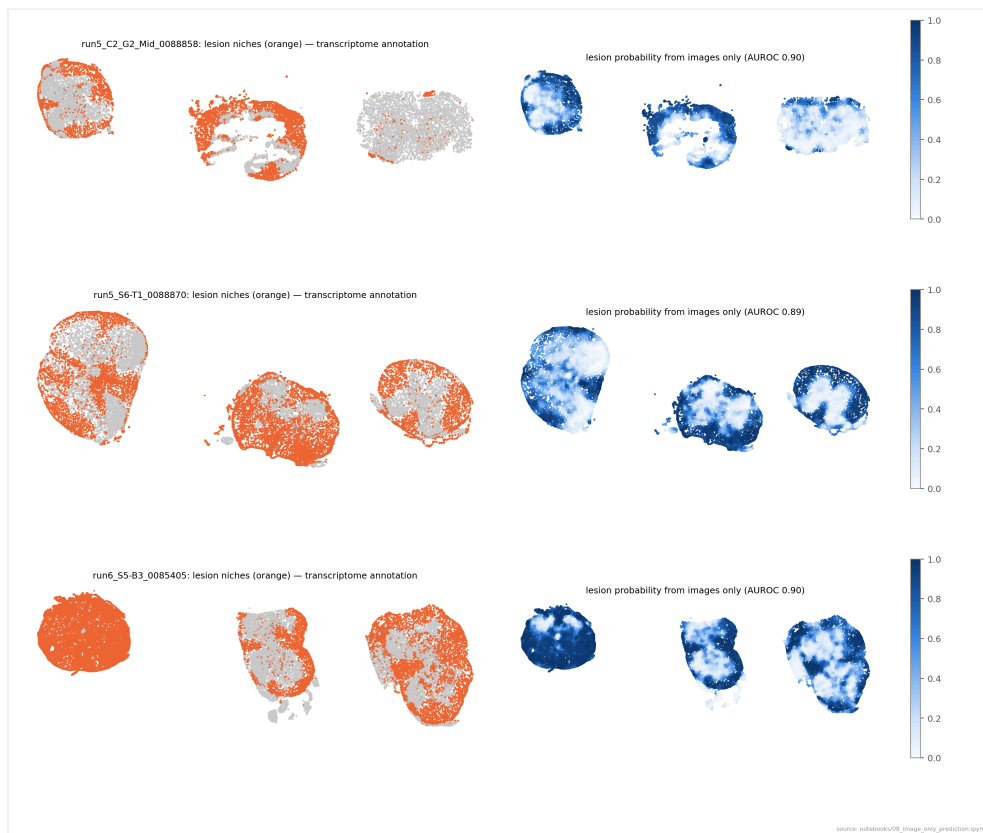


Figure 6 · Lesion maps from images alone. Left: transcriptome-defined lesion niches (orange). Right: lesion probability from image features of held-out animals. Overall AUROC 0.90 (cells alone 0.84; transcriptome neighbourhood 0.92; both 0.93). Top signals: vimentin-channel texture/intensity in the surrounding tissue.

Does it survive a new imaging run?

Yes. Images identify their run with 100 % accuracy (transcriptome 44 %), so a model could lean on run-specific properties. Trained only on run5 animals, the lesion model maps run6 at **AUROC 0.90** (median per animal 0.91, 6 animals); run6 → run5 gives **0.87** (19 animals). Within-run CV gives 0.90. Cell type transfers too (0.42–0.45 vs 0.46). Ranking features within each image doesn't help (0.86–0.87), so per-image normalisation is enough. Caveat: two runs from one batch.

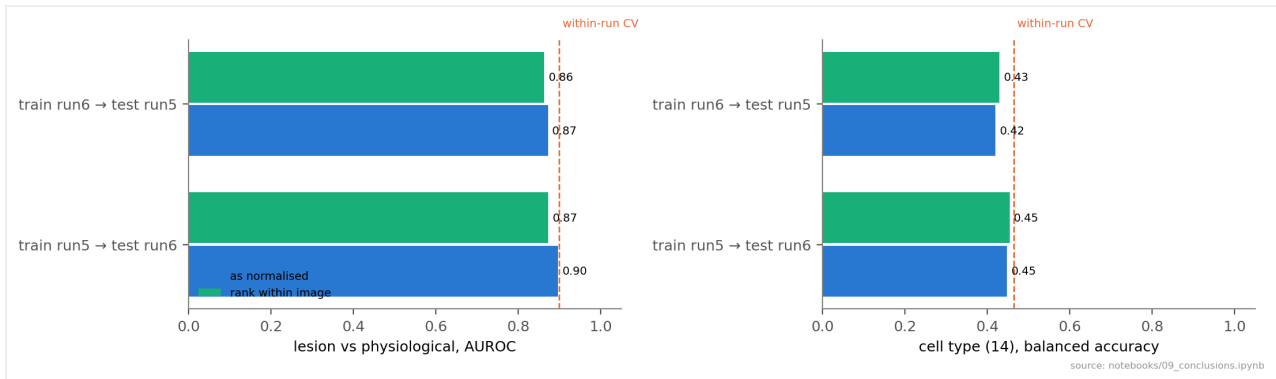


Figure 7 · Cross-run transfer of the image-only lesion (left) and cell-type (right) models; dashed line = within-run cross-validation (notebook 09).

Animal level: clinical score from images p 0.62 (transcriptome 0.79, both 0.76); chronic-phase day of sacrifice 0.52 vs 0.84. Sex is not predictable (negative control). Images are informative about disease but redundant with RNA.

5 Biology only the images show

5.1 White-matter lesions lose neuropil

WM lesion cells have 25 % less surrounding ATP1A1 than physiological WM of the same piece (38 pieces, $p = 8 \times 10^{-6}$); grey matter shows no difference (45 pieces, $p = 0.62$). This is tissue damage that a cell's own transcriptome can't carry. It's highest in ventral/dorsal rim OL niches and not related to clinical score.

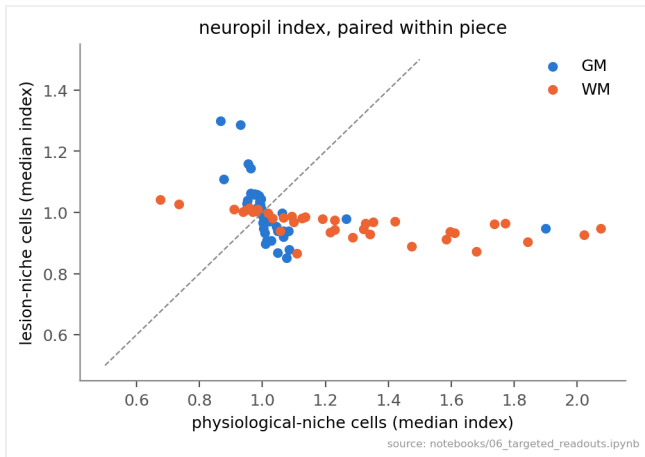


Figure 8a · Neuropil index, lesion vs physiological cells of the same piece.

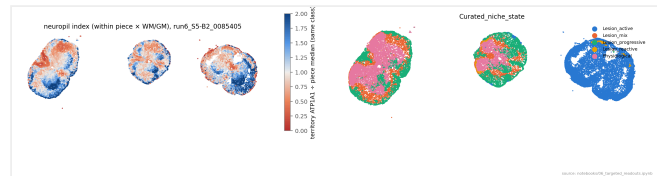


Figure 8b · Neuropil index map next to curated niche states, one section.

5.2 Astrocyte reactive RNA precedes vimentin protein

Protein score (vimentin channel) vs RNA reactivity across 47,062 astrocytes: $p = 0.48$ (*C3* 0.50, *Serping1* 0.39, *Gfap* 0.38; homeostatic *Slc6a11* -0.44). **RNA-only astrocytes** (reactive transcription, quiet vimentin) are 95 % in lesions and peak in active disease (PEAK1 20 %, PEAK2 17 % vs CFA 3.5 %), falling in remission (active vs remission $p = 0.019$). Reactive transcription comes before the protein. *Vim* is not on the panel, so only the stain shows this.

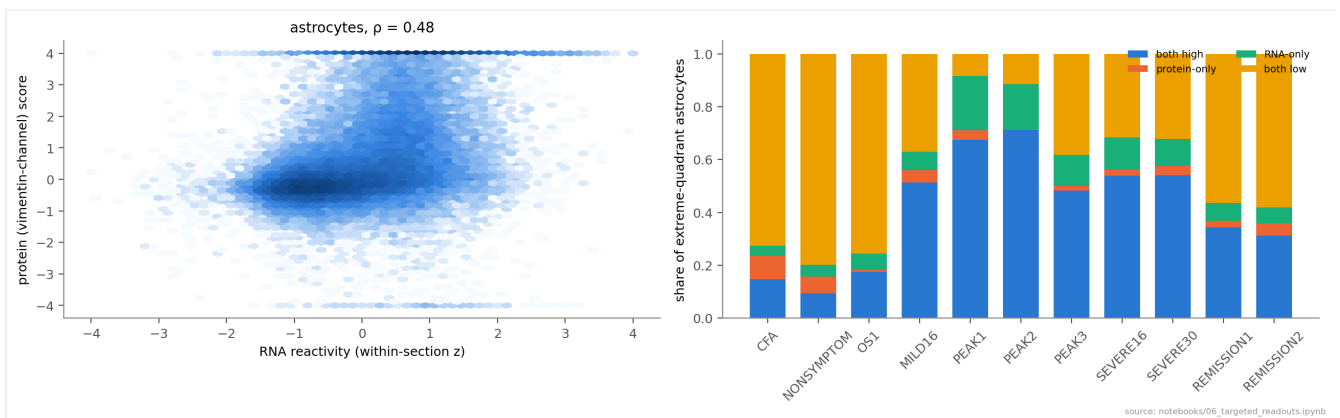


Figure 9 · Astrocyte reactivity, protein vs RNA, and quadrant shares by disease stage (notebook 06).

5.3 18S rRNA: activation

18S rises in activated cells in lesions beyond their RNA content. After adjusting for transcript density: Schwann +0.83, endothelium +0.39, homeostatic astrocytes +0.39 SD. The oligodendrocyte drop (-0.14) disappears (+0.04), so it reflects RNA-content loss.

Withdrawn: "perivascular T cells orient their 18S towards the vessel." A later brightness-matched control (notebook 19) showed T cells point their 18S *more* at an equally bright non-vessel neighbour at the same distance than at the vessel. Measured relative to the cell's own nucleus, nothing vessel-specific remains. The signal is bleed from 18S-bright neighbours plus lopsided outlines of small cells; polarity readouts of small cells next to bright neighbours should not be used.

5.4 Candidate severity marker: 18S texture (needs replication)

Smoother, less punctate 18S in sicker animals in six cell types (p 0.37–0.76). It holds when comparing animals within the same section (p 0.53–0.86) and after adjusting for crowding (0.41–0.81). DAPI texture under the same optics doesn't follow. Possible causes: ribosome redistribution, RNA degradation in inflamed tissue, or animal-level tissue handling.

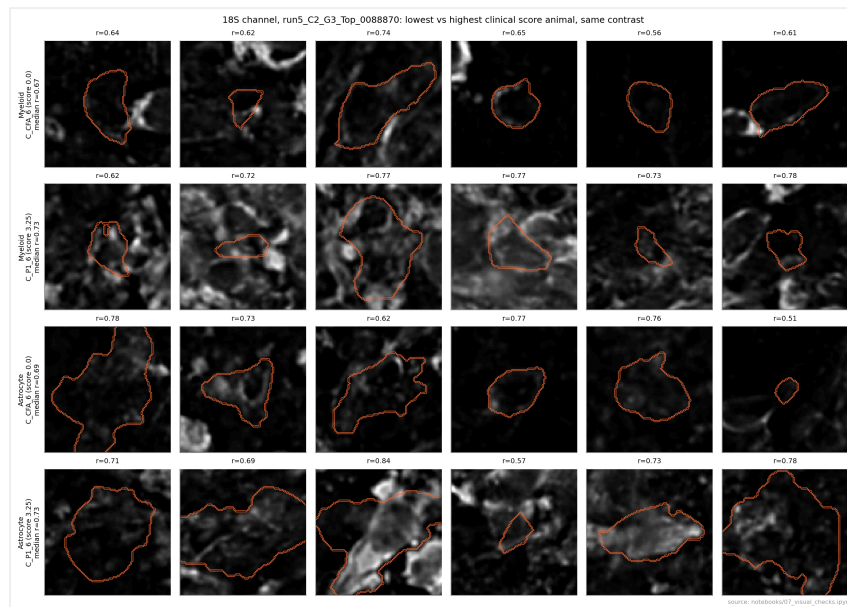


Figure 11 · 18S channel, same section, same contrast: lowest vs highest clinical-score animal (myeloid cells and astrocytes); r = Haralick correlation.

6 How to use the segmentation stains

Use them for	Don't use them for
Vimentin / αSMA / ATP1A1 protein (targets missing from the panel)	CD45 / immune membrane (undetectable in mouse cord)
Tissue context: neuropil loss, anatomy, lesion maps from images	Replacing the transcriptome for lesion identity or severity
Protein-vs-RNA timing (e.g. astrocyte reactivity)	Cell polarity from DAPI or αSMA/Vim (optical bleed)
Per-cell 18S as a ribosome / activation readout (adjusted for transcript density)	Pooling <i>raw</i> feature values across runs (models transfer; raw values carry a run signature)
QC: segmentation method, spill-over-prone neighbourhoods	18S distribution features as biology in 18S-segmented cells

Key numbers

Question	Images	Transcriptome	Both
Cell type, 14 L1 (balanced acc.; chance 0.07)	0.46	—	—
Anatomical region, 10 (balanced acc.; chance 0.10)	0.63	0.68	0.73
Lesion state, 7 (balanced acc.; chance 0.14)	0.54	0.65	0.67
Lesion vs physiological (AUROC)	0.90	0.92	0.93
...trained on one run, tested on the other (AUROC)	0.87–0.90	—	—
Clinical score per animal (p, held-out)	0.62	0.79	0.76
Imaging run identity (acc.)	1.00	0.44	—
Sex (negative control)	0.25	—	—
Image-feature variance explained by covariates + transcriptome (median R ²)			0.11–0.18

Transcriptome inputs for anatomy/lesion are the cell-type composition of the 15 nearest cells (lesion niches are transcriptome-defined, which favours the transcriptome).

Caveats

- Lesion niches are defined from the transcriptome, so image-vs-RNA comparisons for lesions are tilted towards RNA.
- αSMA and vimentin share one channel; a minority of "protein-only" astrocytes sit next to VSMCs (7 % within 10 μm vs 1.5 %).
- 18S drew ~95 % of the cell masks, so 18S distribution features are partly by construction.
- 25 animals, only two CFA control pieces, two imaging runs from one batch; runs 1–3 have no accessible raw images yet.

Next steps

1. **Replicate on runs 1–3** (P drive): 18S texture marker, WM neuropil loss, astrocyte RNA → protein lag, and cross-run transfer of the image models.
2. Separate vimentin from αSMA (VSMC-adjacent subset) and check CD45 antibody mouse reactivity with 10x.
3. Feed the neuropil-loss index and per-cell 18S into the RRMAPP2 lesion/niche analyses as image-derived covariates.